Enumerative Combinatorics — problem 5.4

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Let R_n be the number of ways to place n undistinguishable rooks peacefully on an $n \times n$ chessboard. Moreover, let H_n, Q_n, M_n, D_n be the number of those placings, which are invariant with respect to a half-turn, a quarter-turn, reflection at a diagonal, and reflection at both diagonals. Find formulas for R_n, H_n, Q_n, and find recursions for M_n, D_n.